

Heaps

1. Start with an empty max binomial heap and insert the following keys in order:

21, 11, 34, 42, 9, 37, 15.

When linking two trees of equal degree, make the root with the smaller key the *leftmost child* of the other root. Draw the heap's roots in increasing order of degree.

- (a) Draw the resulting heap. (Feel free to show intermediate ones along the way of course!)

- (b) Now insert 19. Draw the resulting heap and indicate each pair of trees that gets linked. Next, perform `removeMax` on the heap. (Feel free to use the back if space is short!)

2. Prove that a binomial tree of order k has exactly 2^k nodes. (See lecture slides for the full definition and a perhaps helpful figure, and remember that induction is your friend!)